

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	27-06-2026
Team ID	23A51A05G8
Project Name	Software Installation Request Automation in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

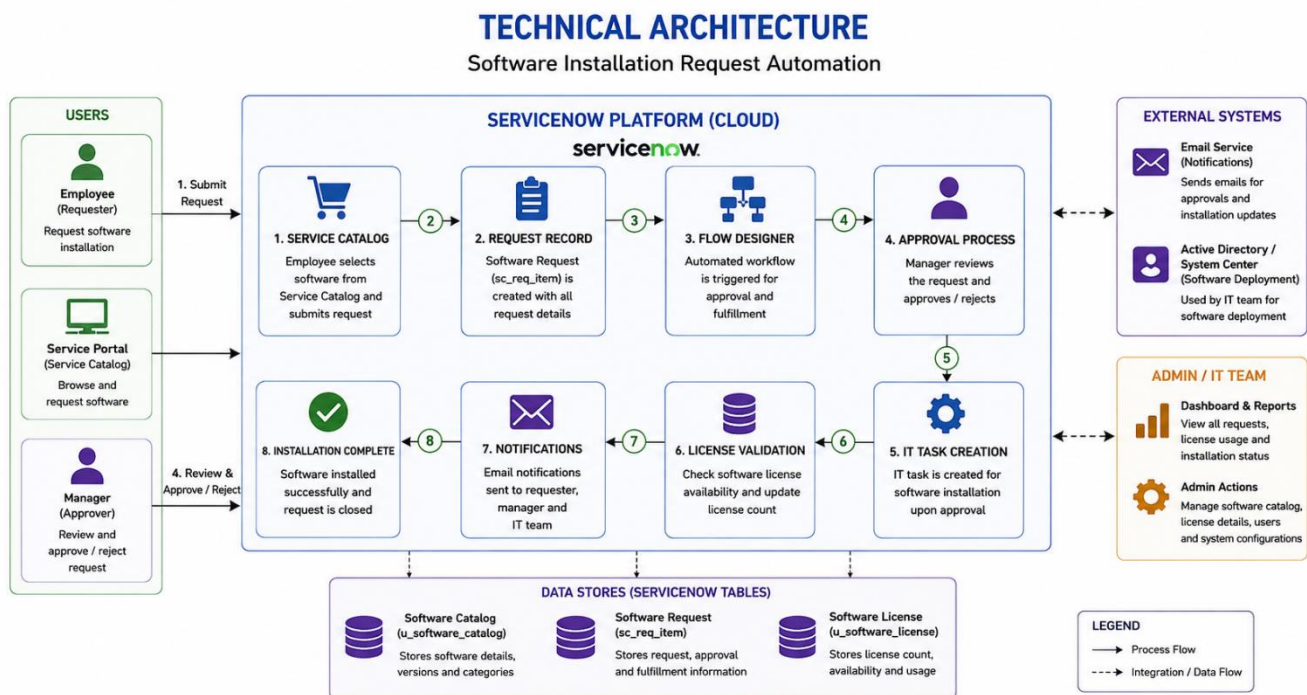


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for employees to submit software installation requests	ServiceNow Service Catalog, Portal UI
2.	Application Logic-1	Client-side form validation, auto-populate manager field, conditional field visibility based on urgency	Catalog Client Scripts (JavaScript)

3.	Application Logic-2	Server-side business rule execution for license check, request date auto-set, and license decrement	Business Rules, GlideRecord API
4.	Application Logic-3	Automated approval workflow, IT task creation, and email notification orchestration	Flow Designer, Business Rules
5.	Database	Stores software catalog master data and all employee software request transaction records	ServiceNow Custom Tables (software_catalog, software_request)
6.	Cloud Database	Cloud-based data storage managed by ServiceNow	ServiceNow Cloud Database
7.	File Storage	Stores request attachments and record history logs	ServiceNow Attachments
8.	External API-1	Email notification service for request lifecycle emails to employees and managers	ServiceNow Notification Engine
9.	External API-2	Approval routing and manager notification service for request approvals	ServiceNow Approval Engine
10.	Reporting & Analytics	Software request trends, license utilization, and approval status reports	ServiceNow Reports & Dashboards
11.	Infrastructure	Application hosting and execution environment	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Client-side scripting and customization used for form validation, manager auto-population, and conditional field logic based on urgency level	JavaScript, GlideRecord API
2.	Security Implementations	Role-based access control for administrators, managers, IT staff, and employees. Authenticated access through ServiceNow platform with secure data storage and transmission	ServiceNow ACLs, User Roles (sw_approver, sw_fulfiller), HTTPS, IAM Controls
3.	Scalable Architecture	ServiceNow's cloud-based multi-tenant architecture supports growing users and software requests without any infrastructure changes	ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform
4.	Availability	High availability through ServiceNow cloud infrastructure with continuous monitoring and managed services ensuring 99.9% uptime	ServiceNow Cloud Infrastructure, Redundant Data Centers

5.	Performance	Fast request processing through server-side Business Rules, optimized GlideRecord queries, and Flow Designer automation reducing manual effort	Business Rules, Flow Designer, ServiceNow Database Optimization
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References:

<https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html>

<https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html>

https://www.servicenow.com/docs/r/api-reference/c_GlideRecordAPI.html

<https://developer.servicenow.com>

<https://www.servicenow.com/community/>